Static IP setup guide

YOU

Oct 11, 03:09 AM

I am trying to set a static IP address with PFSense Firewall and my ASUS AXE11000.. Static IP: 167.248.154.195 Gateway: 167.248.154.193 Subnet Mask: 255.255.255.248 Logs from node: dgarner@Caesar:/var/log$ tail -50 syslog 2025-09-29T21:46:02.120639-05:00 Caesar avahi-daemon[1058]: New relevant interface wlo1.IPv4 for mDNS. 2025-09-29T21:46:02.120765-05:00 Caesar avahi-daemon[1058]: Registering new address record for 192.168.1.69 on wlo1.IPv4. 2025-09-29T21:46:02.120833-05:00 Caesar avahi-daemon[1058]: Registering new address record for 192.168.50.69 on wlo1.IPv4. 2025-09-29T21:46:02.126527-05:00 Caesar avahi-daemon[1058]: Registering new address record for 10.0.0.69 on wlo1.IPv4. 2025-09-29T21:46:02.151561-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1513] dhcp4 (wlo1): state changed new lease, address=10.0.0.150 2025-09-29T21:46:02.153090-05:00 Caesar avahi-daemon[1058]: Registering new address record for 10.0.0.150 on wlo1.IPv4. 2025-09-29T21:46:02.155150-05:00 Caesar systemd-resolved[915]: wlo1: Bus client set search-domain list to: (hack3rpg 2025-09-29T21:46:02.155118-05:00 Caesar systemd-resolved[915]: wlo1: Bus client set default route setting: yes 2025-09-29T21:46:02.157036-05:00 Caesar systemd-resolved[915]: wlo1: Bus client set DNS server list to: 10.0.0.1 2025-09-29T21:46:02.164885-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1643] device (wlo1): state change: ip-config -> ip-check (reason 'none', sys-face-state: 'managed') 2025-09-29T21:46:02.167521-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1673] device (wlo1): state change: ip-check -> secondaries (reason 'none', sys-face-state: 'managed') 2025-09-29T21:46:02.167608-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1675] device (wlo1): state change: secondaries -> activated (reason 'none', sys-face-state: 'managed') 2025-09-29T21:46:02.167886-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1677] manager: NetworkManager state is now CONNECTED_SITE 2025-09-29T21:46:02.168295-05:00 Caesar NetworkManager[1117]: <info> [1759200362.1681] device (wlo1): Activation: successful, device activated. 2025-09-29T21:46:02.177744-05:00 Caesar dhcpcd[93367]: wlo1: soliciting an IPv6 router 2025-09-29T21:46:02.177947-05:00 Caesar dhcpcd[93367]: ps_root_recvmsg: Cannot assign requested address 2025-09-29T21:46:02.278366-05:00 Caesar gvfsd-wsdd[1283]1B: Failed to spawn the wsdd daemon: Failed to execute child process "wsdd" (No such file or directory) 2025-09-29T21:46:02.279284-05:00 Caesar gvfsd-network[7117]: Couldn't create directory monitor on wsdd:///: Error: Automount failed: Failed to spawn the underlying wsdd daemon. 2025-09-29T21:46:02.935028-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 2025-09-29T21:46:05.382682-05:00 Caesar google-chrome.desktop[4047]: [4040:4076:0929/214605.382009:ERROR:google_apis/gcm/engine/connection_factory_impl.cc:434] Failed to connect to MCS endpoint with error -105 2025-09-29T21:46:05.451527-05:00 Caesar google-chrome.desktop[4047]: [4040:4076:0929/214605.451335:ERROR:google_apis/gcm/engine/connection_factory_impl.cc:434] Failed to connect to MCS endpoint with error -105 2025-09-29T21:46:07.383669-05:00 Caesar dhcpcd[93367]: wlo1: leased 10.0.0.151 for 7200 seconds 2025-09-29T21:46:07.384343-05:00 Caesar avahi-daemon[1058]: Registering new address record for 10.0.0.151 on wlo1.IPv4. 2025-09-29T21:46:07.392360-05:00 Caesar dhcpcd[93367]: wlo1: adding route to 10.0.0.0/24 2025-09-29T21:46:07.392666-05:00 Caesar dhcpcd[93367]: wlo1: adding default route via 10.0.0.1 2025-09-29T21:46:07.930021-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-34 noise=9999 txrate=573500 2025-09-29T21:46:09.475473-05:00 Caesar google-chrome.desktop[4047]: [4040:4076:0929/214609.475060:ERROR:google_apis/gcm/engine/connection_factory_impl.cc:434] Failed to connect to MCS endpoint with error -105 2025-09-29T21:46:10.589802-05:00 Caesar systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully. 2025-09-29T21:46:10.950488-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-34 noise=9999 txrate=573500 2025-09-29T21:46:13.948058-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 2025-09-29T21:46:14.183621-05:00 Caesar dhcpcd[93367]: wlo1: no IPv6 Routers available 2025-09-29T21:46:16.952653-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-34 noise=9999 txrate=573500 2025-09-29T21:46:19.973278-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-32 noise=9999 txrate=573500 2025-09-29T21:46:22.975335-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-31 noise=9999 txrate=573500 2025-09-29T21:46:28.999541-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-34 noise=9999 txrate=573500 2025-09-29T21:46:31.998652-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 2025-09-29T21:46:35.003416-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 2025-09-29T21:46:38.025143-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-32 noise=9999 txrate=573500 2025-09-29T21:46:41.038901-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=458800 2025-09-29T21:46:44.037579-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-32 noise=9999 txrate=573500 2025-09-29T21:46:47.045086-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=536600 2025-09-29T21:46:50.084104-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-36 noise=9999 txrate=573500 2025-09-29T21:47:02.084480-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 2025-09-29T21:47:05.085996-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-34 noise=9999 txrate=573500 2025-09-29T21:47:08.108509-05:00 Caesar wpa_supplicant[1119]: wlo1: CTRL-EVENT-SIGNAL-CHANGE above=1 signal=-33 noise=9999 txrate=573500 dgarner@Caesar:/var/log$

AI ASSISTANT

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Got it — you want to assign your public static IP block (from your IP) to your pfSense firewall and use your ASUS GT-AXE11000 as just an access point behind it. Let's step through this carefully.

1. Understand Your Block

You were given:

- **Static IP:** `167.248.154.195`
- **Gateway:** `167.248.154.193`
- **Subnet Mask:** `255.255.255.248`

That subnet mask (`/29`) means you have **8 IP addresses total**:

- `167.248.154.192` → Network
- `167.248.154.193` → Gateway (router at ISP side)
- `167.248.154.194` — `167.248.154.198` → Usable hosts
- `167.248.154.199` → Broadcast

So your usable range is **.194–.198**. You want to use `.195` on pfSense WAN.

2. pfSense WAN Setup

On **pfSense** → **Interfaces** → **WAN**:

1. **IPv4 Configuration Type**: **Static IPv4**
2. **IPv4 Address**: `167.248.154.195`
   Subnet mask: `29` (that's 255.255.255.248)
3. **Gateway**: Add `167.248.154.193` as a new gateway and select it.
4. **DNS**: Set your ISP's or something public (1.1.1.1, 8.8.8.8).

Save & Apply.

3. pfSense LAN Setup

- LAN should stay private (e.g., `10.0.0.1/24`).
- DHCP enabled for your local devices if you want pfSense to hand out addresses.

4. ASUS AXE11000

If you want pfSense to be your **main firewall/router**:

- Put the ASUS into **Access Point / Bridge Mode** (so it only handles Wi-Fi and switches, not routing).
- Connect one of its LAN ports to pfSense LAN.
- Disable DHCP on the ASUS.

5. Check Connectivity

From **pfSense Diagnostics** → **Ping**:

- Ping `167.248.154.193` (your ISP gateway). Should succeed.
- Then try `8.8.8.8` .
- Then try `google.com` .

If the first fails, ISP may need to "bind" your block to your modem/MAC. If the first succeeds but not the second, check your DNS.

6. Your Logs

Right now your node `Caesar` is still picking up DHCP (`10.0.0.150` , `10.0.0.151`) from your ASUS (`10.0.0.1`). That means your ASUS is still doing NAT/DHCP and pfSense is not in control yet.
→ Once you put ASUS in **AP mode** and point WAN on pfSense to the static block, your node should get a `10.0.0.x` from pfSense LAN instead.

✅ Action order:

1. Configure pfSense WAN with static IP block.
2. Switch ASUS to AP mode, uplink it to pfSense LAN.
3. Verify pfSense has Internet via Diagnostics.
4. Verify LAN clients get DHCP from pfSense and reach Internet.

Do you want me to also write out the exact **pfSense "Interfaces → WAN" configuration form values** so you can enter them line-by-line?

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How do I put the router into AP / Bridge mode?

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Perfect — since you've got the **ASUS GT-AXE11000**, here's how you put it into **Access Point (AP) mode** so pfSense is in charge and the ASUS just provides Wi-Fi and switching.

Step-by-Step: AP / Bridge Mode on ASUS AXE11000

1. **Connect to the ASUS admin page**
   - Open a browser and go to http://192.168.50.1 or http://router.asus.com.
   - Log in with your router credentials.
2. **Switch to AP mode**
   - In the left menu, go to **Administration** → **Operation Mode**.
   - You'll see options like:
     - Wireless router mode (default)
     - Access Point (AP) mode
     - Repeater mode
     - Media Bridge
   - Select **Access Point (AP) mode**.
   - Click **Save/Apply**.
3. **Set the LAN IP**
   - During the wizard, the ASUS will ask how to get its LAN IP.
   - Best option: **"Get IP automatically from DHCP"** → pfSense will hand it an address on your LAN.
   - Alternative: Assign it a static IP in your pfSense LAN subnet (e.g., if pfSense LAN = `10.0.0.1` , you can assign ASUS = `10.0.0.2`).
4. **Disable DHCP on ASUS**
   - AP mode should disable DHCP automatically, but double-check:
     - Go to **LAN** → **DHCP Server** and make sure it's **disabled**.
5. **Reconnect cables properly**
   - In **AP mode**, don't use the WAN port.
   - Plug pfSense **LAN** → ASUS **LAN port**.
   - All your devices connect to the ASUS via Wi-Fi or its LAN ports, and pfSense does all the routing/firewall.

Verify

- From a client (like your `Caesar` node), you should now get an IP from pfSense (e.g., `10.0.0.x` if your LAN is 10.0.0.0/24).
- Default gateway should be pfSense LAN (`10.0.0.1`).
- Internet should go out via pfSense WAN static IP (`167.248.154.195`).

⚠️ One tip: the first time you switch to AP mode, you may lose access to the ASUS until you either:

- Check pfSense DHCP leases for its assigned IP, or
- Reconnect directly to it with a cable and scan your LAN.

Do you want me to also give you the **exact LAN IP/DHCP settings I'd recommend** for pfSense + ASUS so you never lose access to the router after it switches to AP mode?